

DLVN Engine

1 1D Hamiltonian

Single-particle lattice Hamiltonian:

$$\hat{H} = \sum_n \alpha_n \hat{c}_n^\dagger \hat{c}_n + \sum_{n \neq m} \beta_{nm} \hat{c}_n^\dagger \hat{c}_m \quad (1)$$

Matrix representation in localized site basis $\{|n\rangle\}$ ($V_{L,R} = 0$):

$$H = \begin{pmatrix} H_L & V_{L,EM} & 0 \\ V_{EM,L} & H_{EM} & V_{EM,R} \\ 0 & V_{R,EM} & H_R \end{pmatrix} \quad (2)$$

2 Liouville-von Neumann Equation with Dissipation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] \quad (3)$$

Phenomenological boundary damping $-\Gamma(\rho - \rho^0)$:

$$\frac{d}{dt} \begin{pmatrix} \rho_{LL} & \rho_{L,EM} & \rho_{LR} \\ \rho_{EM,L} & \rho_{EM,EM} & \rho_{EM,R} \\ \rho_{RL} & \rho_{R,EM} & \rho_{RR} \end{pmatrix} = -\frac{i}{\hbar} [H, \rho] - \Gamma \begin{pmatrix} \rho_{LL} - \rho_L^0 & \frac{1}{2}\rho_{L,EM} & \rho_{LR} \\ \frac{1}{2}\rho_{EM,L} & 0 & \frac{1}{2}\rho_{EM,R} \\ \rho_{RL} & \frac{1}{2}\rho_{R,EM} & \rho_{RR} - \rho_R^0 \end{pmatrix} \quad (4)$$

3 Hamiltonian Diagonalization

Unitary block diagonalization $H_\alpha U_\alpha = U_\alpha \tilde{H}_\alpha$ ($\alpha \in \{L, EM, R\}$):

$$U = U_L \oplus U_{EM} \oplus U_R, \quad \tilde{H} = U^\dagger H U, \quad \tilde{\rho} = U^\dagger \rho U \quad (5)$$

Transformed Hamiltonian block structure:

$$\tilde{H} = \begin{pmatrix} \tilde{H}_L & \tilde{V}_{L,EM} & 0 \\ \tilde{V}_{EM,L} & \tilde{H}_{EM} & \tilde{V}_{EM,R} \\ 0 & \tilde{V}_{R,EM} & \tilde{H}_R \end{pmatrix} \quad (6)$$

Commutator diagonal matrix element evaluation: $[\tilde{H}, \tilde{\rho}]_{ij} = (E_i - E_j) \tilde{\rho}_{ij} = \Omega_{ij} \tilde{\rho}_{ij}$.

4 Equation of Motion

$$\tilde{\rho}_\alpha^0(E_k) = \frac{1}{e^{(E_k - \mu_\alpha)/k_B T_\alpha} + 1}, \quad \mu_L = E_F + \frac{eV_{\text{bias}}}{2}, \quad \mu_R = E_F - \frac{eV_{\text{bias}}}{2} \quad (7)$$

Driven Liouville-von Neumann (DLVN) EOM in state representation:

$$\frac{d\tilde{\rho}}{dt} = -\frac{i}{\hbar} [\tilde{H}, \tilde{\rho}] - \Gamma \begin{pmatrix} \tilde{\rho}_{LL} - \tilde{\rho}_L^0 & \frac{1}{2}\tilde{\rho}_{L,EM} & \tilde{\rho}_{LR} \\ \frac{1}{2}\tilde{\rho}_{EM,L} & 0 & \frac{1}{2}\tilde{\rho}_{EM,R} \\ \tilde{\rho}_{RL} & \frac{1}{2}\tilde{\rho}_{R,EM} & \tilde{\rho}_{RR} - \tilde{\rho}_R^0 \end{pmatrix} \quad (8)$$

5 4th-Order Runge-Kutta Integration

Time propagation $\tilde{\rho}(t + \Delta t) = \tilde{\rho}(t) + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ with pre-allocated slopes:

$$k_1 = f(t, \tilde{\rho}(t)), \quad k_2 = f\left(t + \frac{\Delta t}{2}, \tilde{\rho}(t) + \frac{\Delta t}{2}k_1\right) \quad (9)$$

$$k_3 = f\left(t + \frac{\Delta t}{2}, \tilde{\rho}(t) + \frac{\Delta t}{2}k_2\right), \quad k_4 = f(t + \Delta t, \tilde{\rho}(t) + \Delta tk_3) \quad (10)$$

6 Current Observable

Exact EM -block back-transformation ($\mathcal{O}(N_{EM}^2)$ zero-leakage cost):

$$\rho_{EM,EM} = U_{EM} \tilde{\rho}_{EM,EM} U_{EM}^\dagger \quad (11)$$

Single-bond electrical current ($2 \times$ spin degeneracy factor):

$$I_{n,n+1}(t) = \frac{4e^2}{\hbar} \beta_{n,n+1}^{\text{eV}} \text{Im}[\rho_{n,n+1}(t)] \quad (12)$$

7 4-Terminal 2D Generalization

Multi-terminal block Hamiltonian ($\mathcal{P} = \{L, R, U, D\}$):

$$H = \begin{pmatrix} H_{EM} & V_{EM,L} & V_{EM,R} & V_{EM,U} & V_{EM,D} \\ V_{L,EM} & H_L & 0 & 0 & 0 \\ V_{R,EM} & 0 & H_R & 0 & 0 \\ V_{U,EM} & 0 & 0 & H_U & 0 \\ V_{D,EM} & 0 & 0 & 0 & H_D \end{pmatrix} \quad (13)$$

Multi-lead dissipator $\mathcal{D}[\tilde{\rho}]_{\alpha,\beta}$:

$$\mathcal{D}[\tilde{\rho}]_{\alpha,\beta} = \begin{cases} -\Gamma(\tilde{\rho}_{\alpha,\alpha} - \tilde{\rho}_\alpha^0) & \alpha = \beta \in \mathcal{P} \\ -\frac{1}{2}\Gamma\tilde{\rho}_{\alpha,EM} & \alpha \in \mathcal{P}, \beta = EM \\ -\Gamma\tilde{\rho}_{\alpha,\beta} & \alpha \neq \beta \in \mathcal{P} \\ 0 & \alpha = \beta = EM \end{cases} \quad (14)$$

2D Molecular Grid Hamiltonian ($N_{Mx} \times N_{My}$ lattice) with 1-to-1 edge coupling:

$$H_M = \alpha_M \sum_{i_x, i_y} |i_x, i_y\rangle \langle i_x, i_y| + \beta_M \sum_{\langle (i_x, i_y), (j_x, j_y) \rangle} |i_x, i_y\rangle \langle j_x, j_y| \quad (15)$$